Feature Categorization & Clustering Design

Overview

This document records the clustering methodology and preprocessing decisions for 18 student behavioral features across 8 categories. Each category is designed to produce a categorical label for downstream modeling.

Feature-to-Category Mapping

Feature	Category
`active_days`	Platform Engagement
`total_active_learning_time_seconds`	Platform Engagement
`login_gap_days_mean`	Activity Regularity
`login_gap_days_var`	Activity Regularity
`avg_video_completion_ratio`	Learning Resource Interaction Depth
`videos_interacted`	Learning Resource Interaction Depth
`pause_per_video`	Learning Resource Interaction Depth
`replay_per_video`	Learning Resource Interaction Depth
`avg_scroll_depth_ratio`	Learning Resource Interaction Depth
`avg_hints_per_question`	Help-Seeking
`hint_to_accuracy_ratio`	Help-Seeking
`chatbot_user_messages_per_thread`	AI Interaction Behavior
`chatbot_prompt_effort_chars`	AI Interaction Behavior
`error_recovery_rate`	Reflective Behavior
`average_score_ratio`	High Achievement
`score_variance`	High Achievement
`score_trend`	High Growth
`engagement_trend`	High Growth

Category-Level Clustering Design

1. Platform Engagement

Features: `active_days`, `total_active_learning_time_seconds`

Method: K-Means (k = 3)

Preprocessing: Apply log1p transform to both features to handle right-skewed distributions before clustering.

Rationale: Both features are strictly non-negative and monotonically increasing in meaning. Their relationship is highly linear and positive — students with more active days inevitably accumulate more learning time. Cluster shapes are spherical or elliptical in the transformed space, which matches K-Means' core assumption of minimizing Euclidean distance to centroids. Spectral Clustering provides no advantage here as there is no non-convex cluster geometry.

Class Labels (k = 3):

Class	Label
0	`Disengaged`
1	`Moderate`
2	`Highly Engaged`

2. Activity Regularity

Features: `login_gap_days_mean`, `login_gap_days_var`

Method: K-Means (k = 3)

Preprocessing: Apply log1p transform to `login_gap_days_var`, which is prone to extreme outliers (students who disappear for long periods inflate variance dramatically).

Rationale: These two features jointly define the regularity space. `login_gap_days_mean` captures interval length (frequency), while `login_gap_days_var` captures consistency (stability). The three target clusters — consistent learners (low mean, low var), occasional learners (high mean, low var), and crammers (high mean, high var) — are geometrically convex and linearly separable, making K-Means appropriate. The cluster boundaries are not theoretically defined (unlike High Growth), so a data-driven method is preferred over rule-based splitting.

Class Labels (k = 3):

Class	Label
0	`Crammer`

Class	Label
1	`Moderate`
2	`Consistent Learner`

3. Learning Resource Interaction Depth

Features: `avg_video_completion_ratio`, `videos_interacted`, `pause_per_video`, `replay_per_video`, `avg_scroll_depth_ratio`

Method: Spectral Clustering (k = 3)

Preprocessing: Standardize all features (z-score). Compute affinity matrix using RBF (Gaussian) kernel. Tune the bandwidth parameter `gamma` via nearest-neighbor heuristic.

Rationale: This is the only category where Spectral Clustering is genuinely justified. Five features interact in non-linear ways: high completion rate combined with high pause frequency indicates a deep thinker, whereas high completion rate with zero pauses may indicate passive viewing. These conditional dependencies create non-convex cluster geometries that K-Means cannot capture. Spectral Clustering constructs a similarity graph over students, decomposes the graph Laplacian, and finds clusters in the resulting low-dimensional embedding — correctly handling manifold-like structures in the 5-dimensional feature space.

Class Labels (k = 3):

Class	Label
0	`Surface Learner`
1	`Moderate Learner`
2	`Deep Learner`

4. Help-Seeking

Features: `avg_hints_per_question`, `hint_to_accuracy_ratio`

Method: K-Means (k = 3)

Preprocessing: Clip extreme outliers at the 99th percentile for `avg_hints_per_question`. No log transform required as the range is naturally bounded.

Rationale: These two features form a semantically meaningful 2D quadrant space. The axes directly encode the two dimensions of help-seeking quality: quantity of help requested and effectiveness of that help. K-Means cluster centroids will naturally fall into the quadrant centers, producing interpretable labels. The cluster boundaries are approximately linear, matching K-Means' geometric assumptions. Note: a fourth cluster (`Struggling`: low hints, low accuracy) can be added if k = 4 is preferred for finer granularity.

Class Labels (k = 3):

Class	Label
0	`Independent`
1	`Effective Seeker`
2	`Passive Seeker`

5. AI Interaction Behavior

Features: `chatbot_user_messages_per_thread`, `chatbot_prompt_effort_chars`

Method: K-Means (k = 3)

Preprocessing: Apply log1p transform to `chatbot_prompt_effort_chars`, which is heavily right-skewed (most students write short prompts; a few write very long ones). Students with zero chatbot usage should be assigned the `Non-User` label directly before clustering to avoid polluting the cluster space.

Rationale: After removing non-users, the remaining students occupy a 2D space where message count reflects interaction breadth and prompt length reflects depth of inquiry. Both dimensions are approximately log-normally distributed, and after transformation the clusters become spherical. The three target behaviors (concept explorer, answer-seeker, non-user) are well-separated in this space, making K-Means sufficient. Spectral Clustering would add complexity without improving cluster quality.

Class Labels (k = 3):

Class	Label
0	`Non-User`
1	`Answer-Seeker`
2	`Concept Explorer`

6. Reflective Behavior

Features: `error_recovery_rate`

Method: Percentile Binning (tercile split)

Preprocessing: None required. Handle NaN values (students with no error history) by assigning a separate label `No Data` or imputing with 0.

Rationale: With only one feature, clustering is semantically inappropriate — clustering discovers structure in multi-dimensional space, and there is no structure to discover in a single dimension beyond ordering. Percentile binning at the 33rd and 67th percentiles produces a robust, distribution-agnostic split into three equally populated groups. The

resulting boundaries are stable across random seeds (unlike K-Means) and are trivially interpretable.

```
df['reflective_behavior_label'] = pd.qcut(
    df['error_recovery_rate'],
    q=3,
    labels=['Non-Adaptive', 'Gradually Adaptive', 'Highly Adaptive']
)
```

Class Labels:

Class	Label
0	`Non-Adaptive`
1	`Gradually Adaptive`
2	`Highly Adaptive`

7. High Achievement

Features: `average_score_ratio`, `score_variance`

Method: Gaussian Mixture Model (GMM, n_components = 4)

Preprocessing: No transform needed; both features are bounded ratios. Handle students with only one exam attempt (score_variance = 0 or NaN) as a special case.

Rationale: GMM is preferred over K-Means for two reasons. First, `score_variance` encodes uncertainty — a student near the boundary between `Proficient` and `Inconsistent` may genuinely belong to both profiles, and GMM's soft probabilistic assignments better reflect this ambiguity. Second, the density of high-achieving students (high score, low variance) is typically much smaller than average students, creating unequal cluster sizes and non-uniform covariances. GMM's per-component covariance matrices adapt to these asymmetries, whereas K-Means assumes equal-sized spherical clusters and will systematically misclassify boundary cases.

Class Labels (n_components = 4):

Class	Label
0	`Struggling`
1	`Inconsistent`
2	`Proficient`
3	`Mastery`

8. High Growth

Features: `score_trend`, `engagement_trend`

Method: Rule-based Quadrant Assignment

Preprocessing: Define a threshold `eps` (e.g., 1e-4) below which a slope is treated as zero (stagnant). Handle NaN values by assigning `No Data`.

Rationale: Both features are linear regression slopes whose signs carry direct theoretical meaning — positive means improving, negative means declining. The classification boundary is therefore not empirical but definitional: it is determined by the zero line of each slope. Any data-driven clustering method would find statistically optimal boundaries that may deviate from the theoretically correct zero crossing, producing less interpretable and less theoretically grounded labels. Rule-based assignment is the most honest approach.

```
def classify_growth(row, eps=1e-4):
    s, e = row['score_trend'], row['engagement_trend']
    if abs(s) < eps and abs(e) < eps:
        return 'Stagnant'
    elif s > 0 and e > 0:
        return 'Thriving'
    elif s < 0 and e < 0:
        return 'Declining'
    else:
        return 'Diverging'

df['high_growth_label'] = df.apply(classify_growth, axis=1)
```

Class Labels:

Class	Label
0	`Declining`
1	`Stagnant`
2	`Diverging`
3	`Thriving`

Method Summary

Category	Method	k / Components	Preprocessing
Platform Engagement	K-Means	3	log1p
Activity Regularity	K-Means	3	log1p on variance
Learning Resource Interaction Depth	Spectral Clustering	3	z-score standardization, RBF kernel

Category	Method	k / Components	Preprocessing
Help-Seeking	K-Means	3	clip at 99th percentile
AI Interaction Behavior	K-Means	3	log1p, pre-assign Non-User
Reflective Behavior	Percentile Binning	3 (terciles)	handle NaN separately
High Achievement	GMM	4	handle single-attempt edge case
High Growth	Rule-based	4 (quadrants)	define eps threshold for zero slope